
DeepReasoning: Cognitive Distillation with QLoRA for GSM8K

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Final Project Report

Language Model Architectures - Deep Reasoning Project

Abstract

This report presents DeepReasoning, a project that studies whether a compact language model can be trained to produce structured chain-of-thought style reasoning for grade-school mathematics. The system follows a cognitive distillation pipeline: synthetic reasoning traces are generated with a stronger teacher, a Qwen2.5 model is fine-tuned with QLoRA adapters, and inference-time scaling is evaluated through self-consistency voting. The central finding is nuanced. QLoRA reliably teaches the required output protocol with <thinking>, <reflection>, and <answer> tags, but this formatting gain does not always translate into higher mathematical accuracy. In the primary Qwen2.5-3B study, several initial configurations degraded accuracy relative to the base model, while a carefully tuned configuration reached 81% exact-match accuracy against 72% for the base model on 100 paired GSM8K questions. Additional Qwen2.5-1.5B experiments further show that the adapter improves strict required-format answering but remains below the base model under loose numeric accuracy. These results suggest that cognitive distillation is effective for behavioral alignment and structured reasoning traces, but improving the underlying problem-solving ability requires careful hyperparameter control and evaluation beyond surface-level chain-of-thought formatting.

Keywords QLoRA, GSM8K, cognitive distillation, chain-of-thought, self-consistency, Qwen2.5.

1. Introduction

Large language models can solve many reasoning tasks, but their final answers often provide little insight into the intermediate computation that led to the result. Recent reasoning-oriented systems encourage models to externalize intermediate steps, verify assumptions, and separate final answers from scratch work. The DeepReasoning project investigates a lightweight version of this idea using parameter-efficient fine-tuning rather than full model training.

The project requirement was to implement a three-stage system: a data pipeline for structured reasoning traces, supervised fine-tuning with QLoRA, and inference-time scaling through self-consistency. The target benchmark was GSM8K, a standard collection of grade-school arithmetic word problems. The report focuses on the empirical question that emerged during development: does teaching a model to imitate explicit reasoning improve the final answer, or does it mostly teach the model to imitate a style?

The answer is not absolute. The experiments show that formatting and answer accuracy must be evaluated separately. The adapter can learn to produce the required reasoning structure, yet still lose to the base model under a loose numeric metric. Conversely, with better hyperparameters, the Qwen2.5-3B adapter can improve over the base model on a fixed paired subset. This motivates a careful analysis of both positive and negative results.

Contributions

The main contributions are: (1) an end-to-end QLoRA pipeline for structured mathematical reasoning; (2) a comparative study of sequence length, LoRA rank, learning rate, and inference-time voting; (3) an explicit distinction between strict required-format accuracy and loose numeric answer accuracy; and (4) an interpretation of why stronger base models may leave limited room for teacher-style distillation to improve final answers.

2. Approach

Data and target behavior

The training data follows the GSM8K mathematical reasoning domain. Each example is converted into a structured answer trace with three semantic regions: <thinking> for step-by-step reasoning, <reflection> for verification or self-correction, and <answer> for the final numeric response. This format matches the project requirement of forcing the model to separate reasoning, reflection, and final answer.

Supervised fine-tuning with QLoRA

The base model is loaded in 4-bit NF4 precision and trained through low-rank adapters. Only the adapter parameters are updated, making the approach feasible for limited hardware. The main 3B experiments vary maximum sequence length and LoRA capacity. Later experiments adjust learning rate and number of epochs after observing that the initial high-learning-rate runs harmed answer accuracy.

Inference-time scaling

For self-consistency, the model generates multiple reasoning traces for the same question. Final numeric answers are extracted and a majority vote is used as the consensus prediction. This mechanism tests whether additional inference-time computation improves answer reliability, and it is reported separately from greedy exact-match accuracy.

Metrics

The evaluation separates: greedy exact-match accuracy, strict required-format answer accuracy, loose numeric answer accuracy, valid-format rate, reflection rate, and self-consistency accuracy. This separation is essential because a model may satisfy the requested format without improving the underlying arithmetic answer.

3. Experiments

The primary study uses Qwen2.5-3B-Instruct. This is the model size specified by the project statement and the source of the plots provided during the project discussion. The first ablation compares a 512-token context with two 1024-token configurations. The follow-up configuration reduces the learning rate and trains for one epoch, which becomes the strongest 3B result.

Run	Max len	r	Eval loss	Base	Adapter	Format
len512_r16_a32	512	16	0.301	72%	30%	32%
len1024_r8_a16	1024	8	0.324	72%	29%	31%
len1024_r16_a32	1024	16	0.314	72%	26%	27%

Table 1. Initial Qwen2.5-3B ablation. The high-learning-rate configurations degraded greedy accuracy relative to the base model, even when the adapter learned some of the requested output format.

Additional 1.5B study

The repository also contains a Qwen2.5-1.5B follow-up. This smaller model was used because Qwen2.5-3B already solved many GSM8K questions directly, reducing the margin for improvement through distillation. The 1.5B study is therefore treated as an additional diagnostic experiment rather than the main project result.

4. Results

The initial 3B experiments show that increasing reasoning verbosity is not sufficient. With `max_len=512` the adapter produced less explicit reasoning and stayed close to the base model in the team strict/STAR-style test. With `max_len=1024` the adapter produced longer reasoning traces, but the initial runs performed worse under greedy accuracy. Self-consistency improves both curves, but the adjusted model remains below the base curve in the provided 3B voting plots.

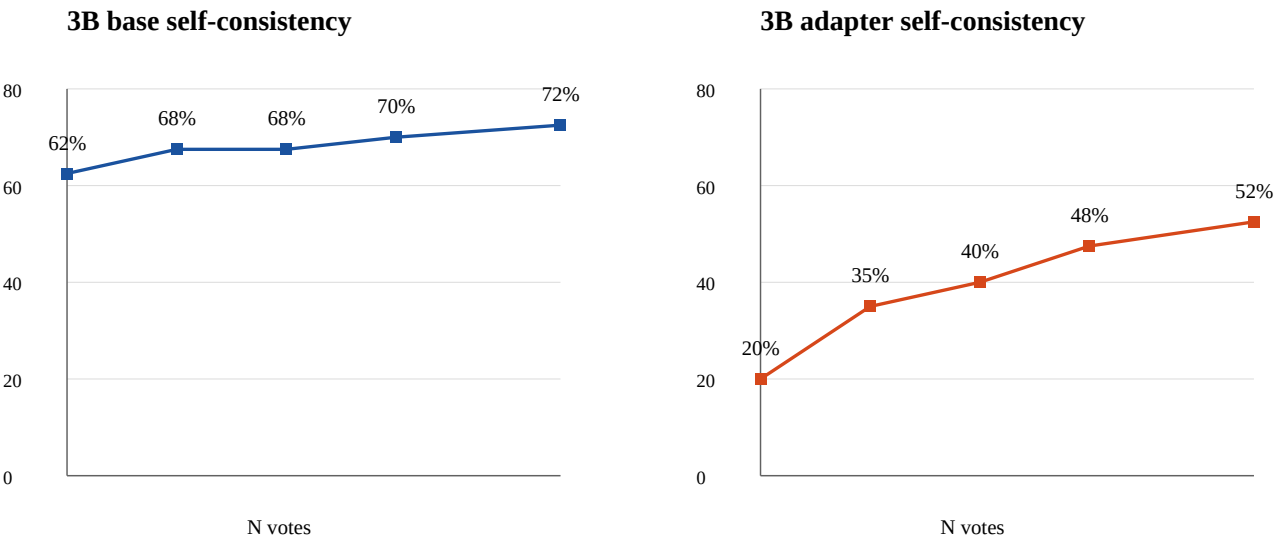
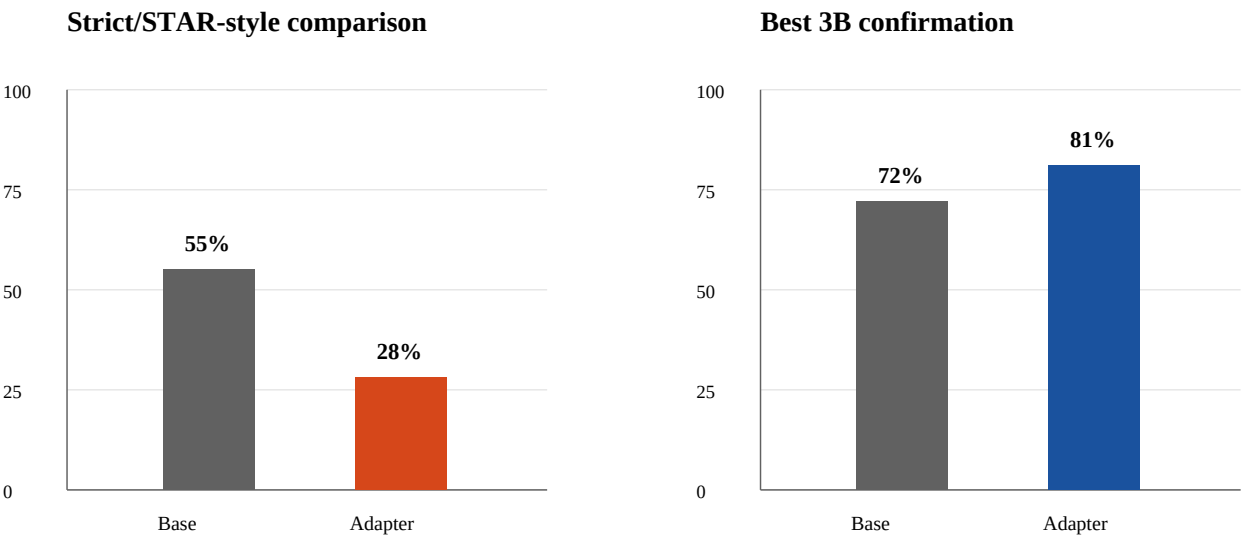


Figure. Self-consistency accuracy versus number of voted reasoning traces for the Qwen2.5-3B experiments provided during the project discussion.



5. Analysis

Why the initial 3B fine-tuning can hurt accuracy

The negative 3B ablation suggests that Qwen2.5-3B already has enough direct problem-solving ability for a large fraction of the benchmark. When the adapter is trained to imitate Gemini-style reasoning, it may allocate capacity to reproducing the teacher's structure rather than improving the final arithmetic decision. A longer context window reduces truncation, but it can also expose the model to more verbose traces, increasing the chance of overfitting to style or introducing unnecessary reasoning noise.

Best 3B result

After reducing the learning rate to $5e-5$ and training for one epoch, the best Qwen2.5-3B adapter reaches 81% exact-match accuracy compared with 72% for the base model on 100 paired GSM8K questions. The paired difference is +9 percentage points with a bootstrap 95% confidence interval of $[-1, 19]$ points. This is the strongest positive result, but the confidence interval remains wide enough that it should be reported as encouraging rather than definitive.

1.5B diagnostic result

For Qwen2.5-1.5B, the adapter obtains 48% strict required-format accuracy versus 0% for the base model. However, under loose numeric accuracy the adapter reaches 49% while the base model reaches 55%. Format compliance is 75% for the adapter and 0% for the base model. This confirms that QLoRA strongly teaches the response contract, but does not automatically improve raw mathematical correctness.

6. Conclusion

DeepReasoning successfully implements the requested project pipeline: cognitive distillation of structured reasoning traces, QLoRA-based supervised fine-tuning, and inference-time self-consistency evaluation. The experiments demonstrate that small adapters can align a model to produce explicit reasoning, reflection, and final-answer tags.

The main lesson is that format learning and reasoning improvement are different outcomes. The adapter often learns to imitate the teacher's reasoning style, but this can fail to improve the final answer and may even reduce accuracy when hyperparameters are poorly chosen. The best Qwen2.5-3B configuration provides an encouraging improvement on a fixed 100-question evaluation, while the Qwen2.5-1.5B diagnostic study shows a clear format gain with a loose-accuracy trade-off.

Future work should evaluate more held-out subsets, include LLM-as-a-judge criteria for logical rigor and self-correction quality, and compare teacher-generated traces against shorter or more answer-focused rationales. The results should be presented as evidence that QLoRA is effective for structured reasoning behavior, but not as proof that longer chain-of-thought imitation necessarily improves mathematical competence.

References

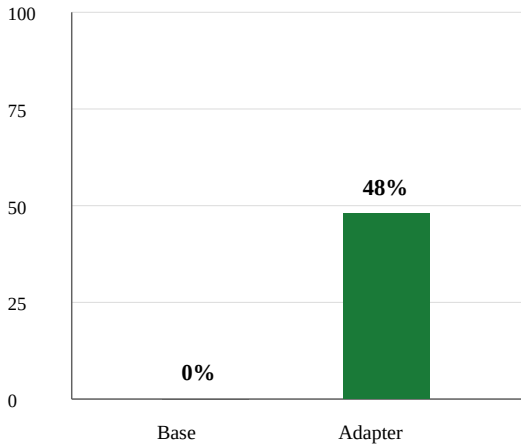
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A. Appendix: Qwen2.5-1.5B Hyperparameter Screen

Run	r	LR	Eval loss	Strict@30	Loose@30	Format@30
r8_a16_lr5e5_e3_es	8	5e-05	0.402	63%	63%	90%
r8_a16_lr2e5_e3_es	8	2e-05	0.456	57%	63%	80%
r16_a32_lr5e5_e3_es	16	5e-05	0.383	53%	57%	80%

The best screened 1.5B adapter is the r=8, alpha=16, learning-rate 5e-5 configuration. The screen supports the main analysis: validation loss, strict format accuracy, and loose numeric accuracy do not always rank configurations in the same order, so all metrics must be reported.

1.5B strict required-format



1.5B loose numeric

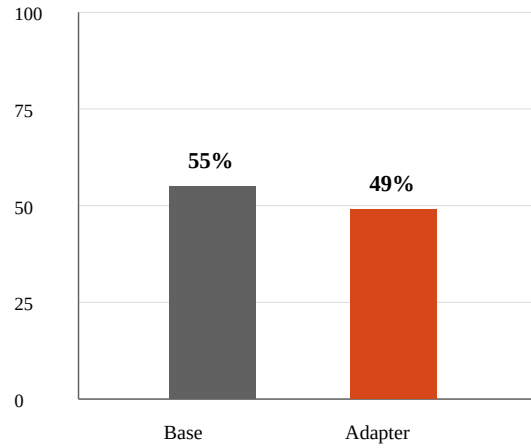


Figure. Qwen2.5-1.5B confirmation: the adapter learns the required response protocol but remains below the base model under loose numeric answer accuracy.